KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

MID-REVIEW PROJECT REPORT

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

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1. Dashboard Overview

This report explains only the KPI cards and plots displayed in the dashboard. It does not include the notebook's general methodology, data-cleaning workflow, or additional EDA plots.

2. KPI Cards – Average Personality Scores

The five KPI cards present the average values for the personality dimensions shown on the dashboard:

Personality Dimension	Displayed Value	Explanation
Neuroticism	0.0043	Highest displayed average among the five traits.
Extraversion	-0.0002	Average value slightly below zero.
Openness	-0.0005	Lowest displayed average among the five traits.
Agreeableness	-0.0002	Average value slightly below zero.
Conscientiousness	-0.0004	Average value slightly below zero.

The KPI cards show Neuroticism as a small positive average, while the other four displayed averages are close to zero and slightly negative.

3. Personality Traits Comparison

The bar chart compares the average values of Neuroticism, Extraversion, Agreeableness, Conscientiousness and Openness. Neuroticism is the only clearly positive bar and is visibly higher than the other four traits. The remaining bars are slightly below the zero line. This chart allows a direct comparison of the five displayed average personality scores.

4. Neuroticism vs Extraversion

The scatter plot places Count of Nscore on the horizontal axis and Average Escore on the vertical axis. The dashboard currently displays a single point labelled “(Blank)” near a count of 50 and an average Escore close to zero. Since only one blank-category point is visible, the plot does not show a spread of observations or a clear trend. A positive or negative relationship cannot be concluded from this display.

5. Neuroticism Across Age Groups by Gender

The line chart compares Neuroticism across six age groups—18–24, 25–34, 35–44, 45–54, 55–64 and 65+—using separate Female and Male lines.

The Female line begins above zero for ages 18–24, moves down towards zero around 35–44, declines through 55–64, and rises again at 65+. The Male line starts around zero, changes only slightly through the middle age groups, and then declines more noticeably in the 55–64 and 65+ groups. At 65+, the displayed Male line is lower than the Female line.

This chart describes differences in the displayed Neuroticism pattern across age groups and gender. It does not establish that age or gender causes changes in personality scores.

6. Key Insights

- Neuroticism has the highest displayed KPI average (0.0043) among the five personality dimensions.
- Extraversion, Openness, Agreeableness and Conscientiousness have small negative averages close to zero.
- The personality comparison bar chart visually reinforces the higher Neuroticism average.
- The Neuroticism vs Extraversion scatter plot shows only one “(Blank)” point, so a relationship cannot be interpreted.
- The age-group line chart shows different Female and Male patterns, with the Male series lower at age 65+.

7. Conclusion

The dashboard provides a compact comparison of five personality-score averages and two visual analyses involving Neuroticism. The KPI cards and bar chart show Neuroticism as the highest displayed average. The scatter plot currently contains only one blank-category point and does not support a relationship interpretation. The age-group line chart displays different Female and Male patterns across the six age categories. This report is limited strictly to the information visible in the dashboard.